

termvar, x			
terminals	::=		
		$\longrightarrow$	
		$\rightarrow$	
		$\Rightarrow$	
		λ	
		$\mapsto$	
		$\vdash$	
		$\emptyset$	
		$\times$	
		$<:$	
		$\langle$	
		$\rangle$	
Γ	::=		
		$\emptyset$	<i>contexts:</i> <i>empty context</i>
		$\Gamma, x : T$	<i>term variable binding</i>
formula	::=		
		judgement	
		$x : T \in \Gamma$	
t	::=		
		(t)	M
		x	<i>variable</i>
		$\lambda x : T. t$	bind x in t <i>abstraction</i>
		$t_1 t_2$	<i>application</i>
		$[x \mapsto t']t$	M
v	::=		
		$\lambda x : T. t$	bind x in t <i>values:</i> <i>abstraction value</i>
T, S, U	::=		
		$T \rightarrow T'$	<i>types:</i> <i>type of functions</i>
		(T)	M
		Top	<i>maximum type</i>
Jop	::=		
		$t \longrightarrow t'$	<i>Evaluation</i>
Jtype	::=		
		$\Gamma \vdash t : T$	<i>Typing</i>
		$T_1 <: T_2$	<i>Subtyping</i>
judgement	::=		
		Jop	
		Jtype	
user_syntax	::=		
		termvar	

	terminals
	Γ
	formula
	t
	v
	T

$t \longrightarrow t'$ *Evaluation*

$$\frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \quad \text{E_APP1}$$

$$\frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \quad \text{E_APP2}$$

$$\frac{}{(\lambda x : T.t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{E_APPABS}$$

$\Gamma \vdash t : T$ *Typing*

$$\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \quad \text{T_VAR}$$

$$\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2} \quad \text{T_ABS}$$

$$\frac{\begin{array}{c} \Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \\ \Gamma \vdash t_2 : T_{11} \end{array}}{\Gamma \vdash t_1 t_2 : T_{12}} \quad \text{T_APP}$$

$$\frac{\begin{array}{c} \Gamma \vdash t : S \\ S <: T \end{array}}{\Gamma \vdash t : T} \quad \text{T_SUB}$$

$T_1 <: T_2$ *Subtyping*

$$\frac{}{S <: S} \quad \text{S_REFL}$$

$$\frac{\begin{array}{c} S <: U \\ U <: T \end{array}}{S <: T} \quad \text{S_TRANS}$$

$$\frac{\begin{array}{c} T_1 <: S_1 \\ S_2 <: T_2 \end{array}}{S_1 \rightarrow S_2 <: T_1 \rightarrow T_2} \quad \text{S_ARROW}$$

$$\frac{}{S <: \text{Top}} \quad \text{S_TOP}$$

Definition rules: 11 good 0 bad
Definition rule clauses: 23 good 0 bad